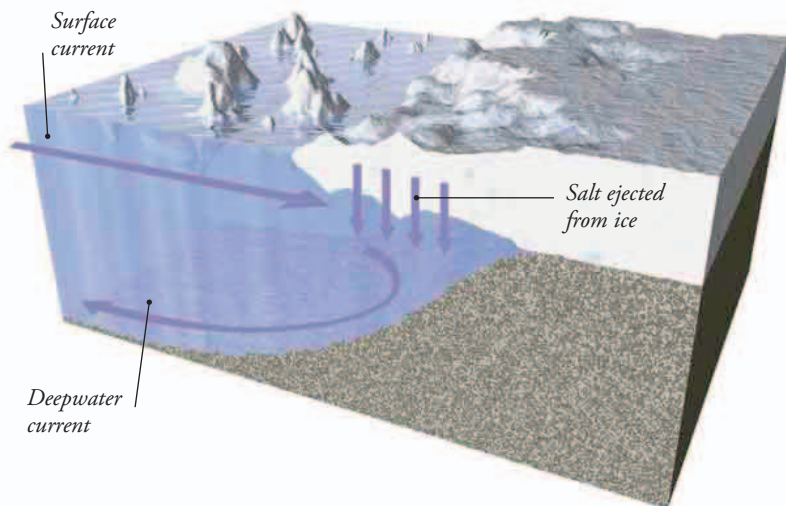
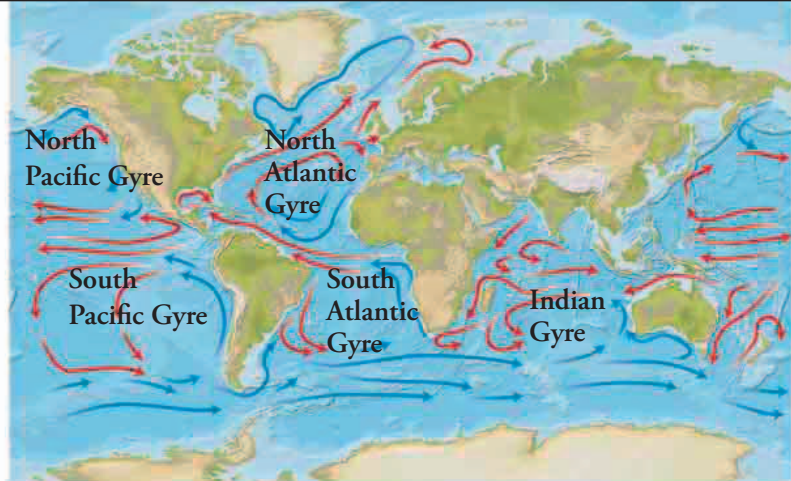


SURFACE CURRENTS

Winds over oceans usually blow steadily from a particular direction. These prevailing winds drive the surface water of the oceans in five huge gyres (right) that swirl clockwise in the Northern Hemisphere, and counterclockwise in the Southern Hemisphere.

They carry warm tropical water (red arrows) toward the poles, and cold polar water (blue arrows) toward the tropics.

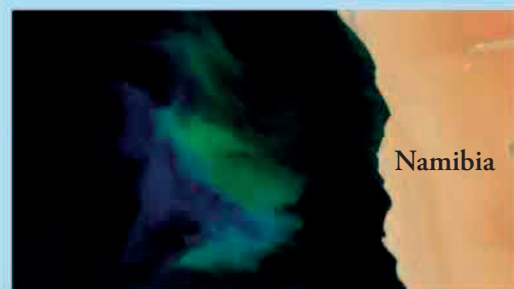


DEEPWATER CURRENTS

In cold oceans, the water surface freezes and ejects salt. The salt and icy cold make the water below the ice denser and heavier, so it sinks toward the ocean floor. This cold, dense ocean water then flows very slowly beneath warmer, less dense water, and gradually mixes with it until it reappears at the surface.

UPWELLING ZONES

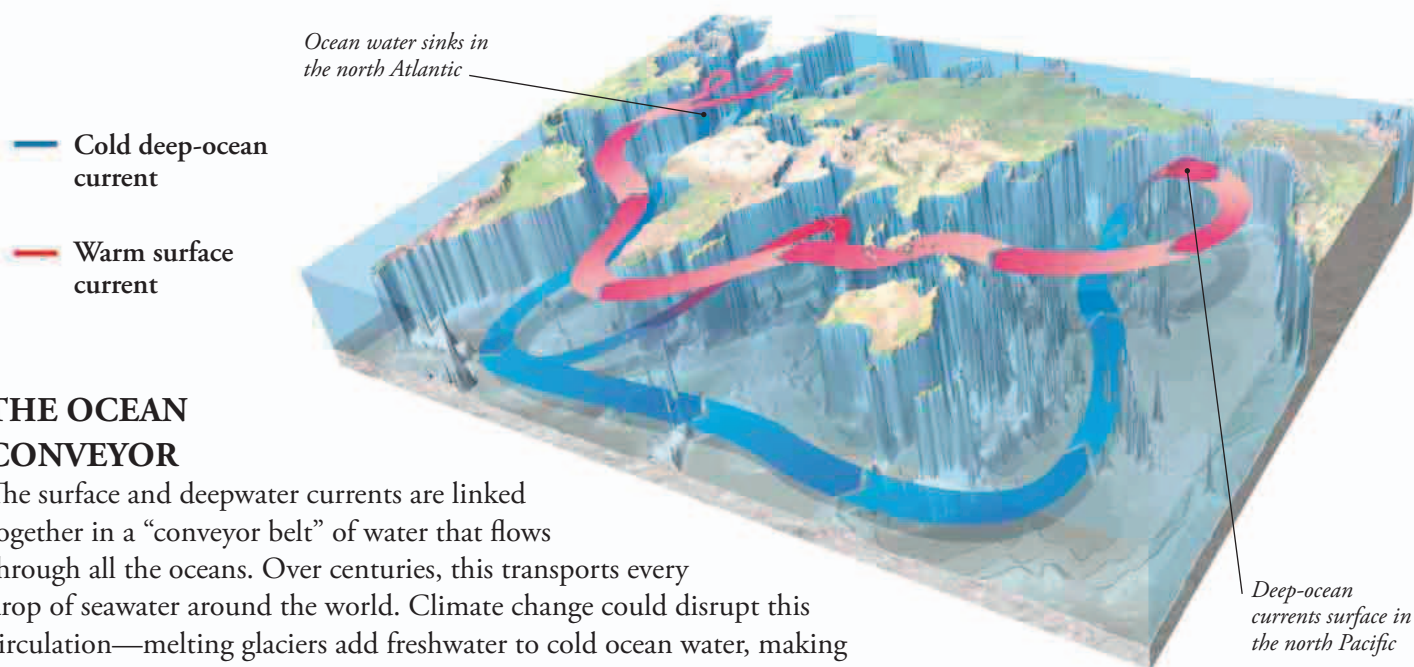
When surface water is driven away from coasts by the wind, deeper water rises to take its place. It comes up loaded with minerals, fueling the growth of plankton—as seen here off southwest Africa. The plankton supports dense populations of animals, so these upwelling zones teem with marine life.



Namibia

▲ PLANKTON BLOOM

Blooms of plankton, such as this one off Namibia, can be many miles across.



THE OCEAN CONVEYOR

The surface and deepwater currents are linked together in a “conveyor belt” of water that flows through all the oceans. Over centuries, this transports every drop of seawater around the world. Climate change could disrupt this circulation—melting glaciers add freshwater to cold ocean water, making it less salty, and less likely to sink and drive deepwater currents.